

A HALF-TURN SYMMETRIC OCTAGON OBSTRUCTING THE STANDARD PSEUDO-TRIANGULATION ROAD MAP

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ABSTRACT. An official AIM workshop summary already records a negative answer to the symmetry question for the pseudo-triangulation algorithm, but gives no explicit witness. We give a rational simple nonconvex octagon for which every legal first step of the standard pointed-pseudo-triangulation road map breaks the original labelled half-turn symmetry. Every convex-hull edge belongs to a half-turn orbit of size two. The road map deletes exactly one hull edge from a pointed pseudo-triangulation, leaving its symmetric partner as a rigid bar. Any half-turn-preserving motion must keep the two paired distances equal, so the deleted distance is constant as well. All edge lengths of the original infinitesimally rigid completion are then fixed, and no nontrivial first motion is possible. This does not obstruct the distinct canonical symmetry-preserving expansive motion of Connelly, Demaine, and Rote, nor a modified pseudo-triangulation method that deletes an entire symmetry orbit.

1. QUESTION AND STATUS

AIM Problem 13 asks whether the pseudo-triangulation algorithm for the carpenter’s-rule problem gives an unfolding that preserves the symmetries of the initial linkage. We use “the pseudo-triangulation algorithm” in its published standard sense: complete the polygon to a pointed pseudo-triangulation, delete one eligible convex-hull edge, and follow the resulting one-degree-of-freedom expansive mechanism.

The AIM workshop summary, publicly archived by September 18, 2015, states that this question is false and separately notes that the construction of Connelly, Demaine, and Rote does preserve symmetry. It supplies neither a witness nor a proof. Accordingly, the negative status is not new: the contribution of this note is an explicit rational witness and a self-contained obstruction argument for the standard road map. No priority claim is made for the negative answer itself.

2. THE SYMMETRIC OCTAGON

Let $P = (p_0, \dots, p_7)$ be the polygon with vertices, in cyclic order,

$$\begin{aligned} p_0 &= (2, 1), & p_1 &= (0, \tfrac{1}{2}), \\ p_2 &= (-2, 1), & p_3 &= (-1, 0), \\ p_4 &= (-2, -1), & p_5 &= (0, -\tfrac{1}{2}), \\ p_6 &= (2, -1), & p_7 &= (1, 0). \end{aligned}$$

The half-turn

$$h(p_i) = p_{i+4} = -p_i$$

is a labelled symmetry of the polygonal linkage.

Lemma 2.1. *The polygon P is simple and nonconvex. Its convex hull has vertices*

$$p_0, p_2, p_4, p_6,$$

and none of its four hull edges is a polygon edge. The hull-edge orbits under h are

$$p_0p_2 \longleftrightarrow p_4p_6, \quad p_2p_4 \longleftrightarrow p_6p_0.$$

Proof. Direct determinant calculations show that the signed turns alternate between -2 and $3/2$. Thus the polygon is nonconvex and has no collinear consecutive vertices. Checking the twenty pairs of nonincident edges shows that no pair intersects. The four extreme vertices are exactly $(\pm 2, \pm 1)$, while the remaining four vertices lie strictly inside their convex hull. The displayed half-turn sends each hull edge to the opposite one, and no hull edge joins consecutive polygon vertices. □

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3. THE STANDARD ROAD-MAP STEP

A pointed pseudo-triangulation on n vertices is a maximal planar pointed graph and has $2n - 3$ edges. A simple polygon, regarded as a noncrossing pointed edge set, can be completed to such a graph. The standard road-map convexification step chooses a pointed-pseudo-triangulation completion T , deletes one convex-hull edge that is not a polygon edge, pins a remaining edge, and follows the resulting one-degree-of-freedom expansive mechanism until the next alignment event.

For the octagon in Lemma 2.1, every completion T contains the eight polygon edges and all four convex-hull edges. These twelve edges are distinct. Since T has

$$2 \cdot 8 - 3 = 13$$

edges, the completion contains only one additional edge. The argument below does not depend on that edge or on the choice of completion.

4. THE MISSING-EDGE OBSTRUCTION

Theorem 4.1. *No legal first step of the standard pointed-pseudo-triangulation road map for P preserves its labelled half-turn symmetry.*

Proof. Fix an arbitrary pointed-pseudo-triangulation completion T , and let $e = p_a p_b$ be the hull edge deleted in the first road-map step. By Lemma 2.1,

$$h(e) = p_{a+4} p_{b+4} \neq e.$$

The partner $h(e)$ is another mandatory hull edge of T , and it remains a rigid bar in the mechanism $T - e$.

Suppose that a motion $q_i(t)$ of $T - e$ preserved the original labelled half-turn. We allow both its center and the whole framework to move. Thus there would be a point $c(t)$ such that

$$q_{i+4}(t) = 2c(t) - q_i(t) \quad (i \in \mathbb{Z}/8\mathbb{Z}).$$

It follows that

$$q_{a+4}(t) - q_{b+4}(t) = -(q_a(t) - q_b(t)),$$

and hence

$$\|q_{a+4}(t) - q_{b+4}(t)\| = \|q_a(t) - q_b(t)\|.$$

The left-hand side is the length of the retained bar $h(e)$, so it is constant. The missing distance e is therefore constant as well. Consequently every edge length of the full pointed pseudo-triangulation T is constant during the motion.

Pointed pseudo-triangulations are infinitesimally rigid and hence locally rigid. The congruence orbit of the initial realization is therefore both closed and locally open inside any continuous path preserving all edge lengths of T . Since the parameter interval is connected, the whole path remains in that congruence orbit. After the road map pins an edge, even this ambient rigid-body motion is removed.

The first road-map segment is required to be the nontrivial expansive motion of the one-degree-of-freedom mechanism $T - e$. It therefore cannot preserve the half-turn. Since T and e were arbitrary, every legal first step breaks the symmetry. $\square$

Remark 4.2. The infinitesimal form is even shorter. Orient the unique mechanism flex so that the deleted hull distance has positive derivative. Half-turn symmetry would make this derivative equal to that of the retained partner bar, which is zero.

5. SCOPE

Remark 5.1. The theorem concerns the standard road map in which exactly one hull edge is deleted at each step. It does not rule out a modified method that deletes an entire symmetry orbit and analyzes a higher-dimensional mechanism. It also does not assert that no symmetric pointed pseudo-triangulation exists.

There is no contradiction with the symmetry-preserving carpenter's-rule motion of Connelly, Demaine, and Rote. Their construction selects the unique solution of a global optimization problem; applying a symmetry to the solution gives the same solution by uniqueness. That canonical motion is a different mechanism from the finite sequence of one-edge-deletion pseudo-triangulation motions considered here.

APPENDIX A. CHECKLIST FOR VERIFICATION

- (1) Check polygon simplicity and the alternating signed turns.
- (2) Recompute the four hull vertices and verify that no hull edge is a polygon edge.
- (3) Verify the two size-two half-turn orbits of hull edges.
- (4) Check that every pointed-pseudo-triangulation completion contains all polygon and hull edges and has thirteen edges.
- (5) For each possible deleted hull edge, verify that its half-turn partner remains a rigid bar.
- (6) Use the moving-center half-turn equation to prove equality of the two paired distances.
- (7) Apply local rigidity of the full completion and keep the standard road map separate from other expansive motions.

REFERENCES

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